



Fondamenti di Analisi dei Dati

from **data analysis** to **predictive techniques**

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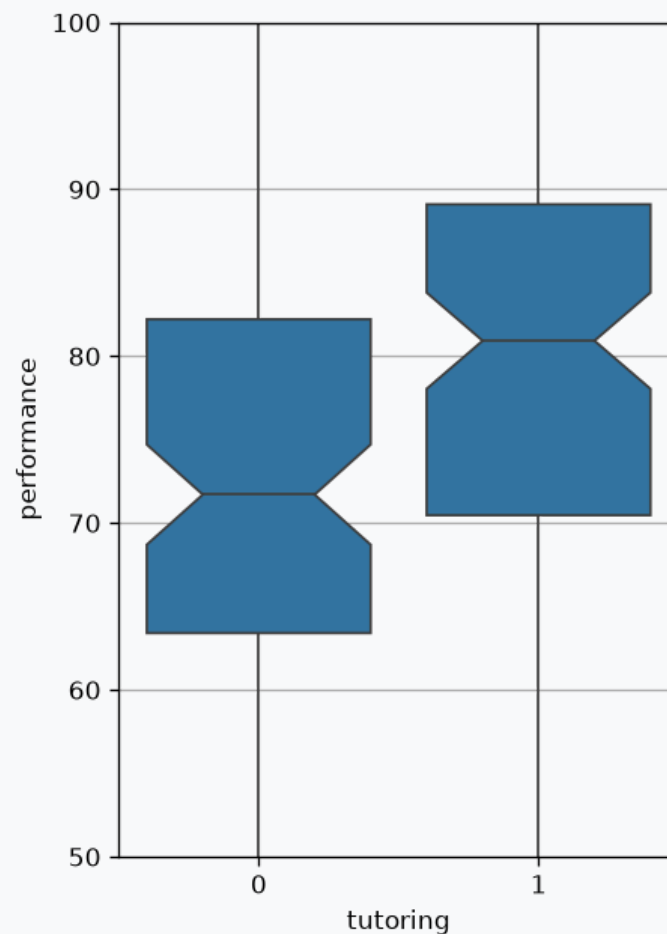
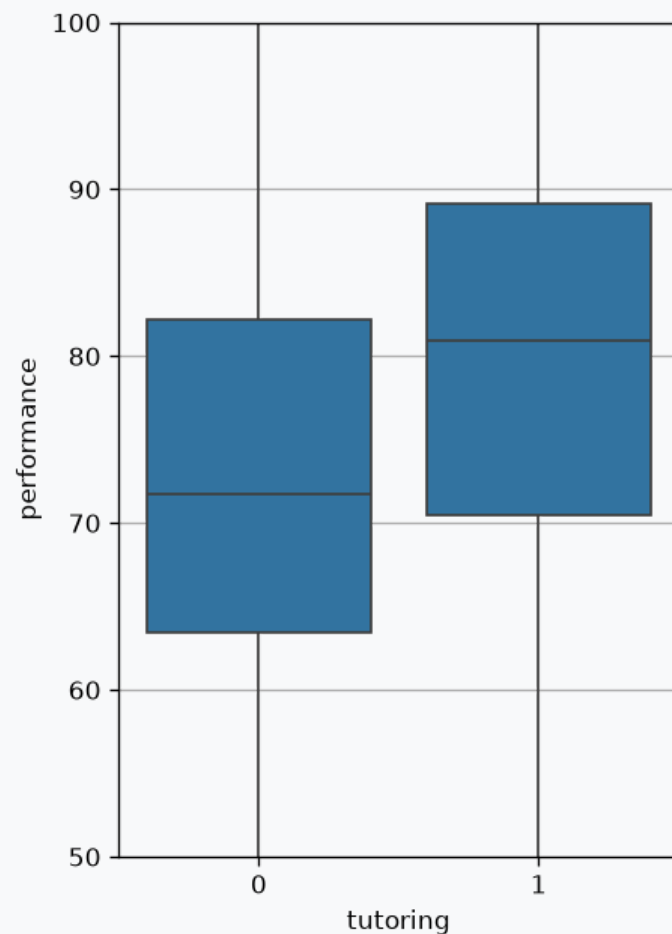
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Causal Analysis

What has to be true before the word "causes" can be used at all

Tutoring vs Performance

A school wants to see if a new teaching method is effective. They ask students if they want to **voluntarily enroll in a tutoring program**. After one year, they compare the average performance of students who participated with those who did not.



The comparison tells us **tutoring** seems to **significantly improve** performance.

We took a statistics class, so we check. The **notches** mark the confidence intervals of the medians: they do not overlap.

T-statistic: 4.087
p-value: 6.36e-05

It looks like the tutoring program is a great success. **Is it?**



Measuring Causation: Potential Outcomes

We are seeing a correlation. Is it also causation? To assess a causal relationship we would ideally **take the same student and let them attend and not attend the tutoring at the same time** — impossible, and unethical even if it were possible.

The notation

- T_i : the **treatment**, 1 if student i attended the tutoring, 0 otherwise. The two groups are the **treatment group** and the **control group**.
- Y_i : the **outcome**, here the performance of student i .
- $Y_i(t)$: the outcome of unit i when the treatment takes value t .

For a given unit we **either observe $Y_i(0)$ or $Y_i(1)$, never both**.

Potential outcomes

The value we *would* observe if we could intervene on T_i and change it. It is not real and cannot be observed, but it is a useful theoretical object — also called a **counterfactual**, because it answers "what if...".

With potential outcomes, the **individual treatment effect** is $Y_i(1) - Y_i(0)$, and averaging over the population gives:

$$ATE = E[Y(1) - Y(0)]$$

$$ATT = E[Y(1) - Y(0) | T = 1]$$



The Bias Arising from Correlation

Without potential outcomes we cannot compute ATE or ATT . What we **can** measure is the difference between two distinct groups:

$$E[Y(1)|T = 1] - E[Y(0)|T = 0]$$

This is what a **correlation** measures: the change in the average of Y across two groups with different characteristics. Adding and subtracting the counterfactual $E[Y(0)|T = 1]$:

$$\underbrace{E[Y(1)|T = 1] - E[Y(0)|T = 0]}_{\text{what we measure}} = \underbrace{ATT}_{\text{what we want}} + \underbrace{E[Y(0)|T = 1] - E[Y(0)|T = 0]}_{\text{bias}}$$

- The bias term compares the **untreated outcome** of the two groups. It is zero only when the groups would have behaved the same without the treatment — when they are **comparable in everything except the treatment**.

If students who choose the tutoring are the more motivated ones, they would have scored higher anyway, and the difference we measured is not an effect of tutoring.



Randomized Controlled Trials

The most robust way to remove this bias is a **randomized experiment**: individuals are randomly assigned to the treatment or the control group, so that apart from the treatment the two groups are indistinguishable.

Random assignment makes the potential outcomes conditionally independent given the treatment:

$$Y(0) \perp Y(1) \mid T$$

If $Y(0) \perp Y(1) \mid T$, then **the treatment is the only thing generating a difference** between the outcome in the treated and in the control group.

It follows that $E[Y(0) \mid T = 0] = E[Y(0) \mid T = 1] = E[Y(0)]$, and likewise for $Y(1)$, hence:

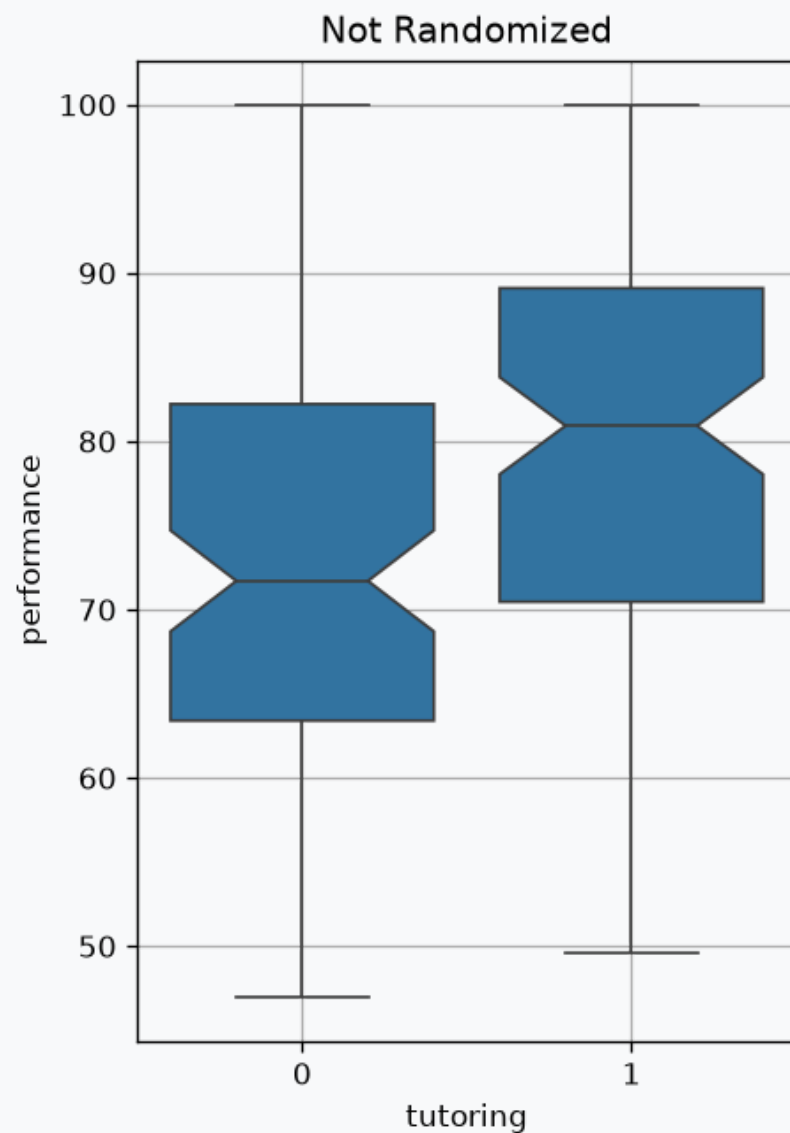
$$E[Y(1) \mid T = 1] - E[Y(0) \mid T = 0] = E[Y(1)] - E[Y(0)] = ATE$$

In this case, **correlation would be causation**.



Randomizing Tutoring

Let's repeat the experiment with random assignment. (A word of caution: this is not always possible — it would not be very polite to force students into a programme.)



T-statistic: -1.133
p-value: 0.259

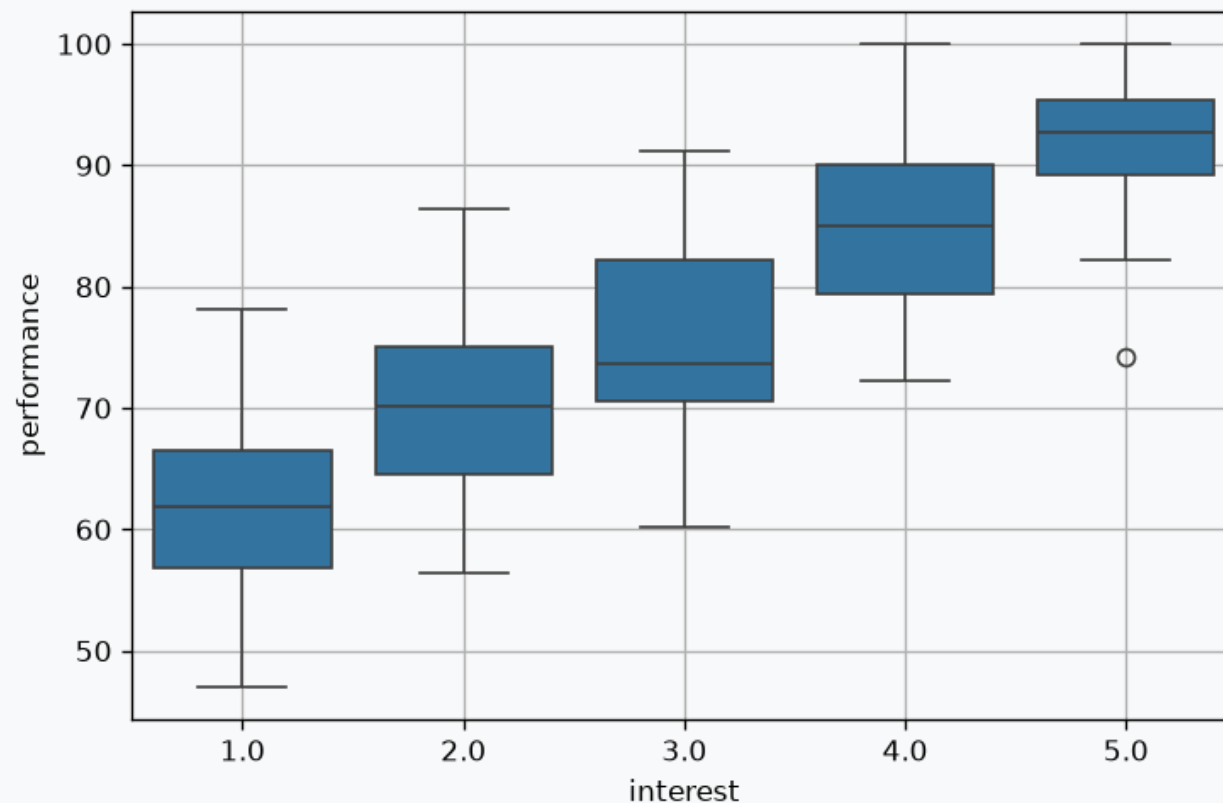
We now see a **less dramatic, and actually negative** difference, and the p-value is large: it is not statistically significant.

By randomising we got rid of **selection bias**. Students who volunteered were probably more motivated and keen to study; the effect of tutoring itself is very small.

Causal Effect and Observational Studies

The gold standard is the randomized controlled trial, but **it is not always feasible or ethical**. Consider estimating the effect of smoking on a disease: we would have to select subjects randomly and ask one group to smoke.

In these cases we can only use **observational data** — and we know the process is subject to bias. The rest of this lecture is about a framework that says **where the bias comes from**, so that it can be removed by **controlling** on the right variable.



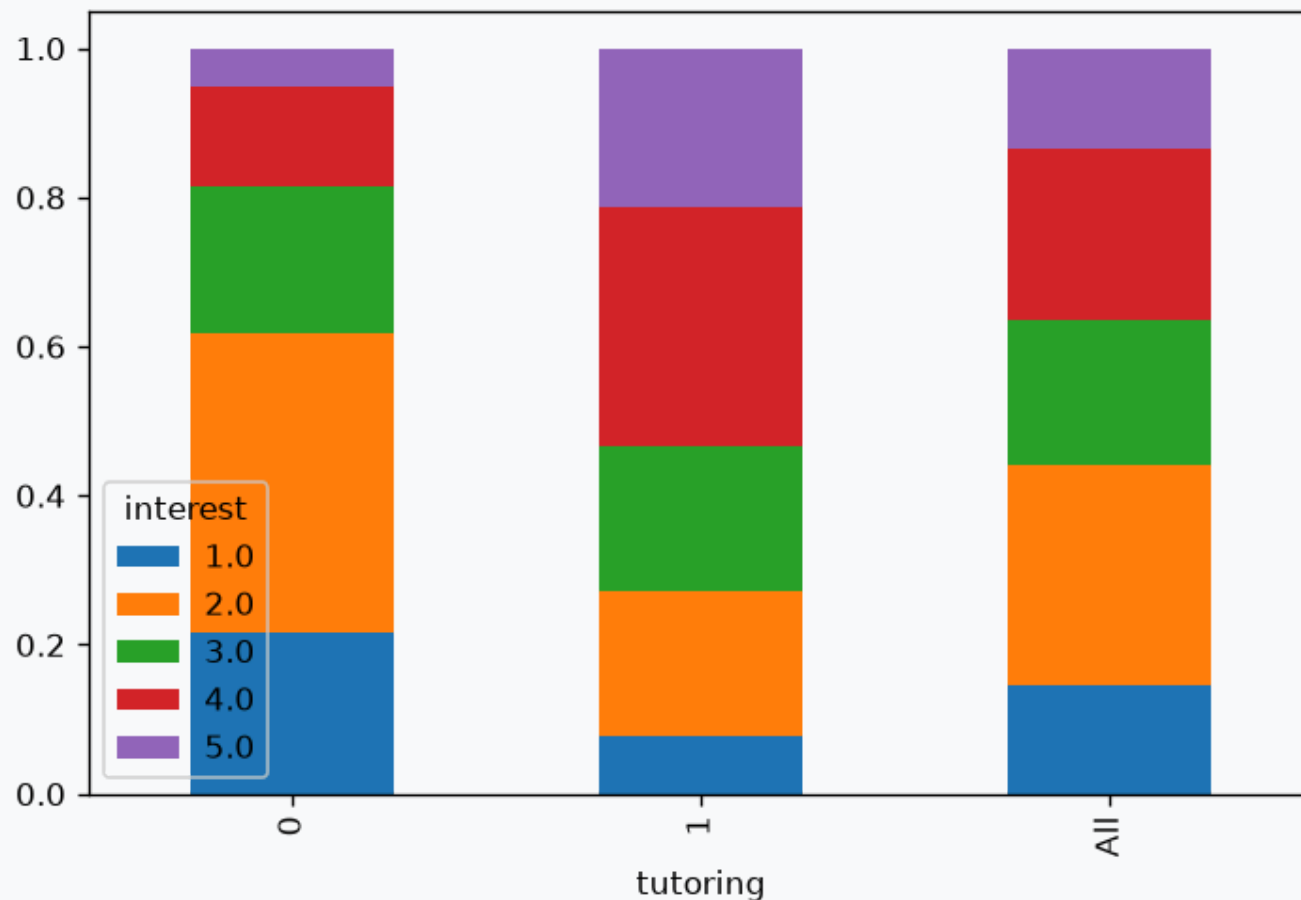
Back to the students: we cannot randomise, so we ask them how interested they are in studying, from 1 to 5.

These are **the same observations as the first experiment**, with one variable added.



Controlling for **interest**

If we split the data by **interest** and compare performance **within each group**, the picture changes:



Within students of the same interest, the difference between the tutored and the untutored **collapses**.

This is what "controlling for a variable" means: comparing like with like, and letting the association that survives be the causal one.

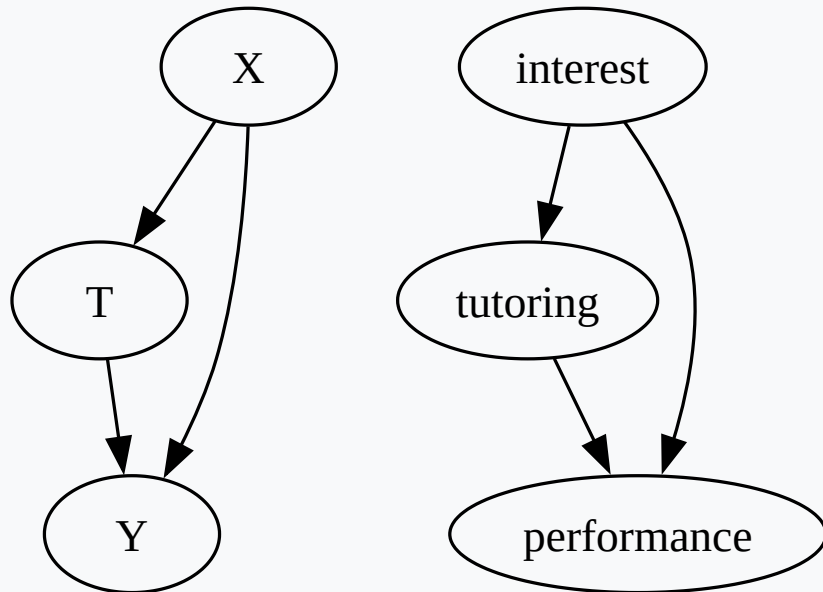
But how do we know **which** variables to control for? Controlling for the wrong one makes things worse.



Graphical Causal Models

How do we identify the variables we should control for?

A **graphical causal model** is a **directed acyclic graph** formalising cause-effect relationships: each node is a random variable, and an edge $A \rightarrow B$ says that A **causes** B .

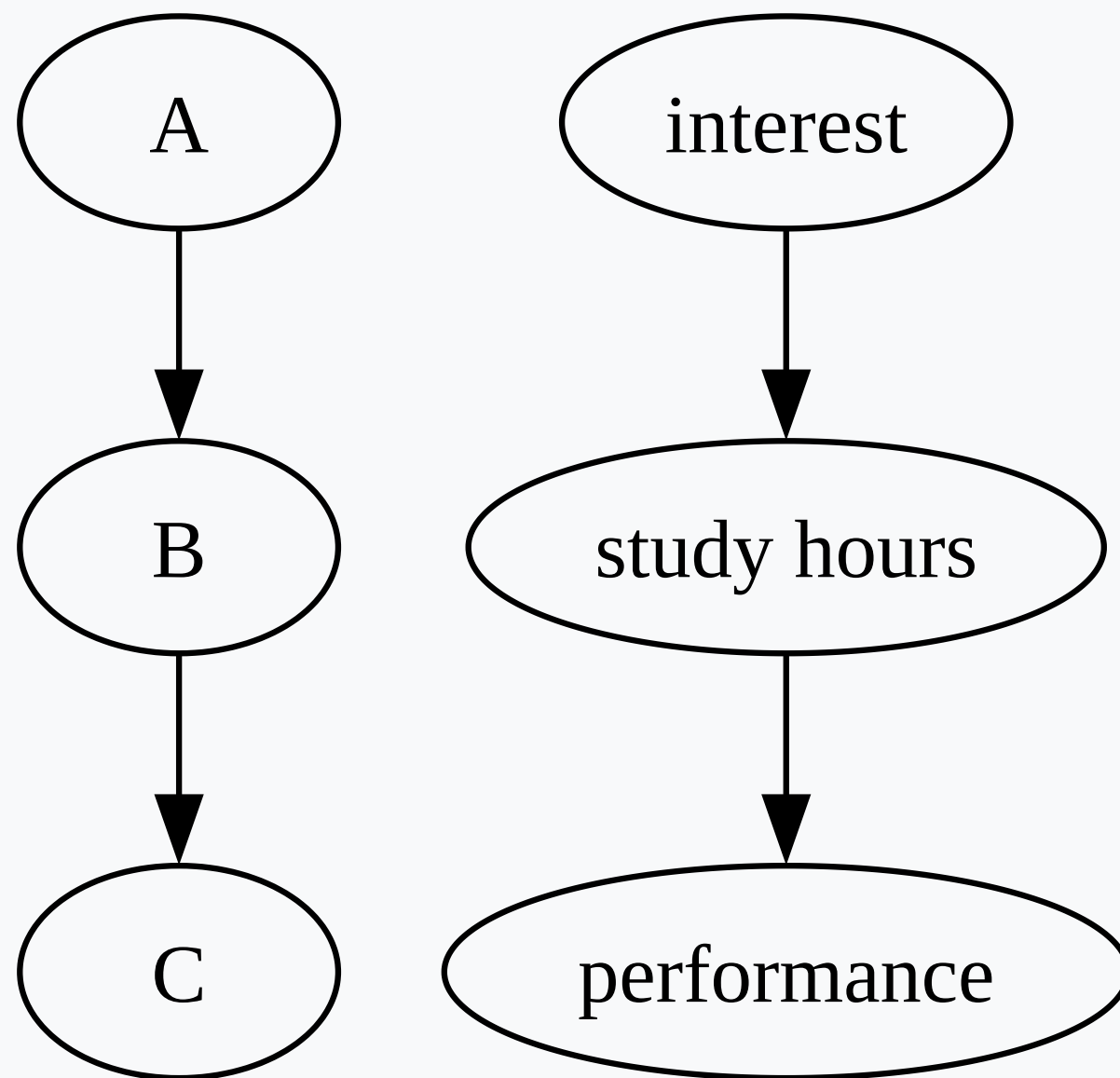


We expect **interest** (X) to influence both **performance** (Y) and **tutoring** (T). From the graph we can identify **interest** as a source of bias.

- Graphical models are **designed by the analyst**, who encodes their own beliefs about how the world works. They come from a Bayesian view: first state the beliefs, then use a framework to draw conclusions from them.



Chains



`interest` causes `study hours`, which in turn causes `performance`. All three pairs are associated:

- if a person has great interest, they study a lot and perform well;
- if a person performs well, we infer they study a lot and are interested.

If we **control for** B (`study hours`), `interest` and `performance` are **no longer associated**:

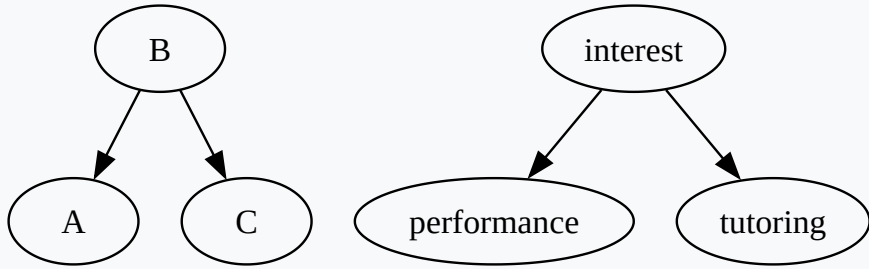
Among people who study the same number of hours, a person with good performance could have great interest or little interest — we cannot tell them apart.

$$A \not\perp C \quad \text{but} \quad A \perp C \mid B$$

Conditioning on B **blocks the dependency**: there is a causal effect between A and C , but we cannot see it any more.



Forks



A variable causes two others. Here B (`interest`) causes both A (`performance`) and C (`tutoring`) — the example we started from.

B is the **common cause**. Although we observe correlations between any pair, the graph says there is **no causal relationship between A and C** .

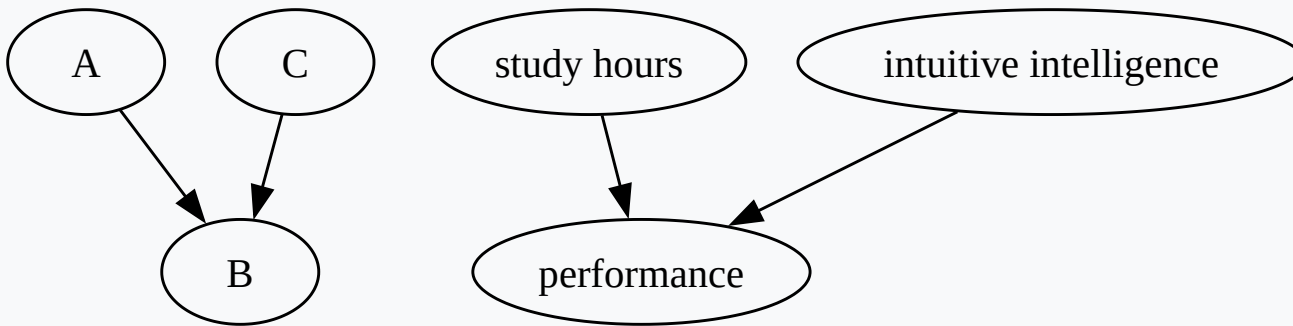
The fork creates a **backdoor path**: dependence flows through B to get from A to C . Conditioning on B closes it, and the association vanishes:

$$A \perp C \mid B$$

Which is exactly what we saw when we compared students of the same interest.



Colliders



The opposite of a fork: a variable is **caused by** two others. `performance` is caused by `study hours` and by `intuitive intelligence`.

A collider **blocks dependency by default**: knowing that a person studies a lot says nothing about their intuitive intelligence.

$$A \perp C$$

But if we **condition on the collider**, the two causes become associated — and the association is not a causal effect:

Among students with good performance, one who does not study much must be intuitively intelligent. Knowing one cause **explains away** the other.

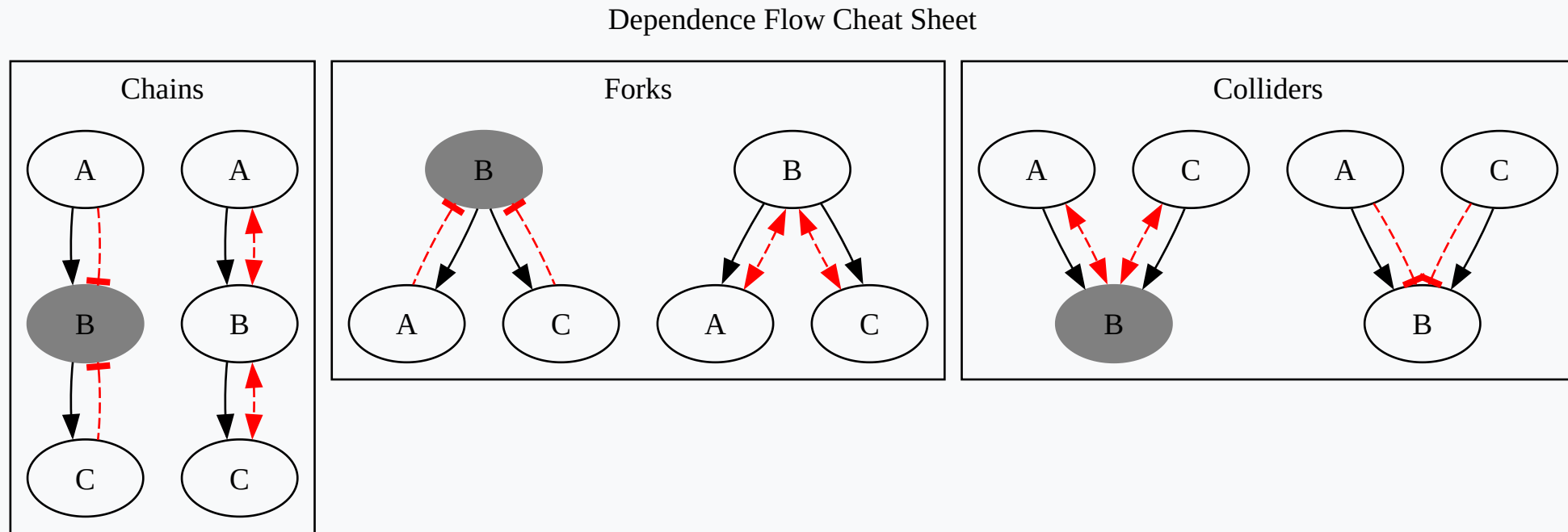
$$A \not\perp C \mid B$$

This is why controlling for a variable "just to be safe" is not safe.



Dependence Flow Rules

Two nodes are dependent when **an unblocked path exists between them**. Combining the three structures gives the cheat sheet:



Solid black arrows are cause-effect relations; dashed red arrows show how dependence flows. A grey node is a node we are conditioning on, and a bar across an arrow means the flow is blocked.

Blocked Paths, and d-Separation

Two nodes are independent when **every path between them is blocked**. A path between A and B is blocked if:

It contains a non-collider that has been conditioned on

A chain or a fork through a grey node stops the flow.

It contains a collider that has not been conditioned on

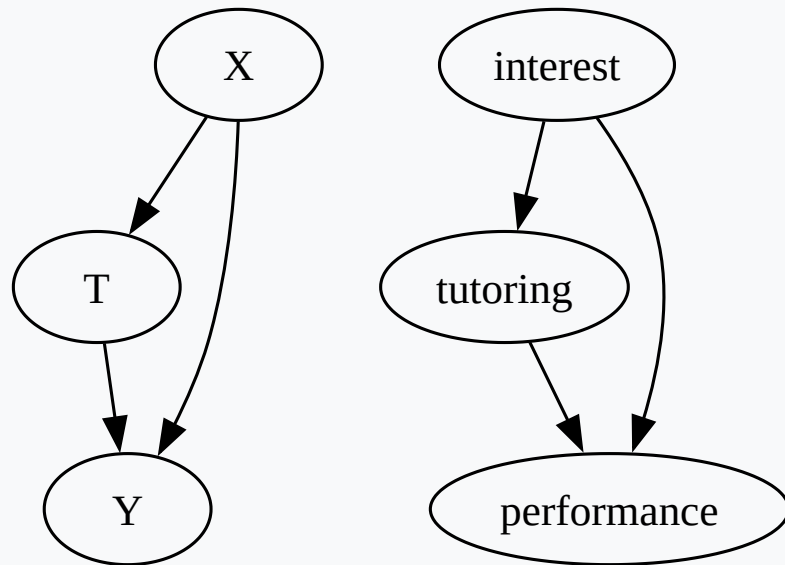
...and none of whose descendants has been conditioned on either.

- Deciding whether two nodes are independent given a set of others is a **mechanical procedure** — trace every path, block it or not according to the three rules — and it is worth doing on paper a few times; the exercises at the end of the chapter ask you to.
In practice you will rarely do it by hand: what matters for an analysis is **recognising confounding and selection bias**, which is what comes next.



Confounding Bias

Confounding bias happens when **the treatment and the outcome have a common cause**.



The common cause X (**interest**) is a **confounder**: it mixes up the effect of **tutoring** on **performance** with its own. If we measure the correlation between treatment and outcome we get a large number, and we do not know how much of it is the confounder.

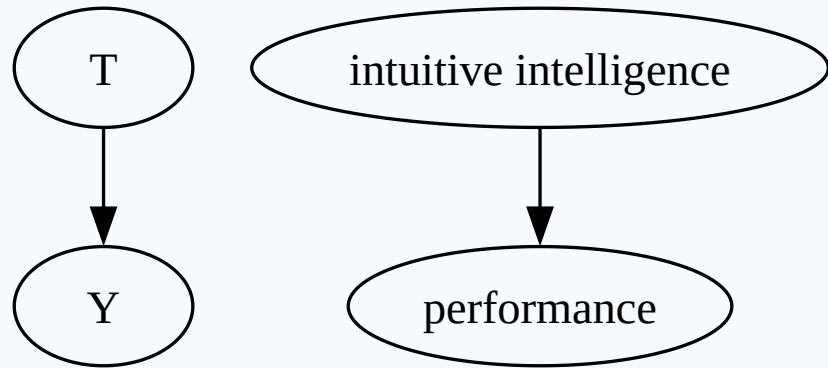
To measure the **direct effect** we must **close the backdoor path** by conditioning on the common cause — that is, comparing subjects with the same level of interest:

$$Y(0) \perp Y(1) \mid T, X$$



Selection Bias: Controlling for a Common Effect

While confounding comes from **not** conditioning on a common cause, selection bias comes from **conditioning on too much**.



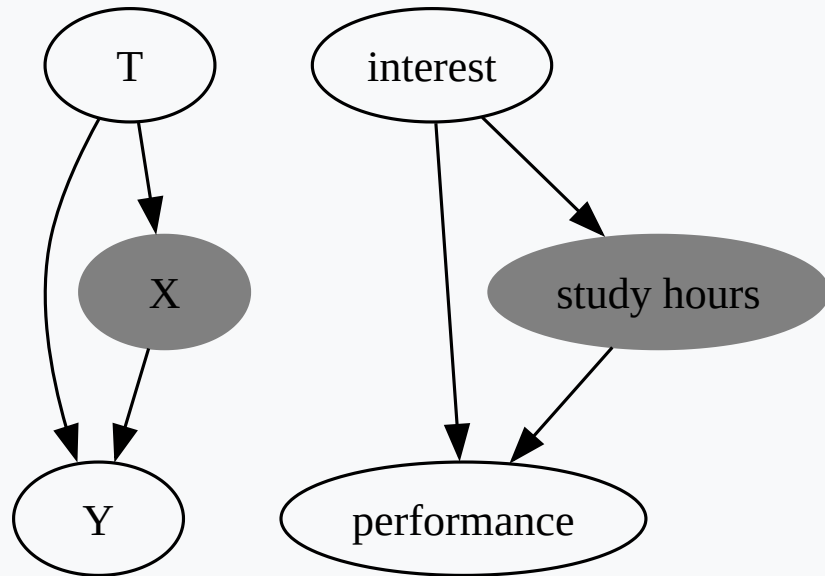
We want the effect of **intuitive intelligence** on **performance**. Our data also has a **problem solving** score, so "just to be sure" we condition on it.

problem solving is a **common effect** of the two. Among people who solve problems well, knowing that someone performs well **explains away** their intuitive intelligence — so we see a correlation that is not a causal effect.

By conditioning on the common effect we opened a backdoor path. The same happens if we condition on a *descendant* of it.



Selection Bias: Controlling on a Mediator



We want the effect of `interest` on `performance`, and this time we could randomise interest. "Just to be sure", we condition on `study hours`.

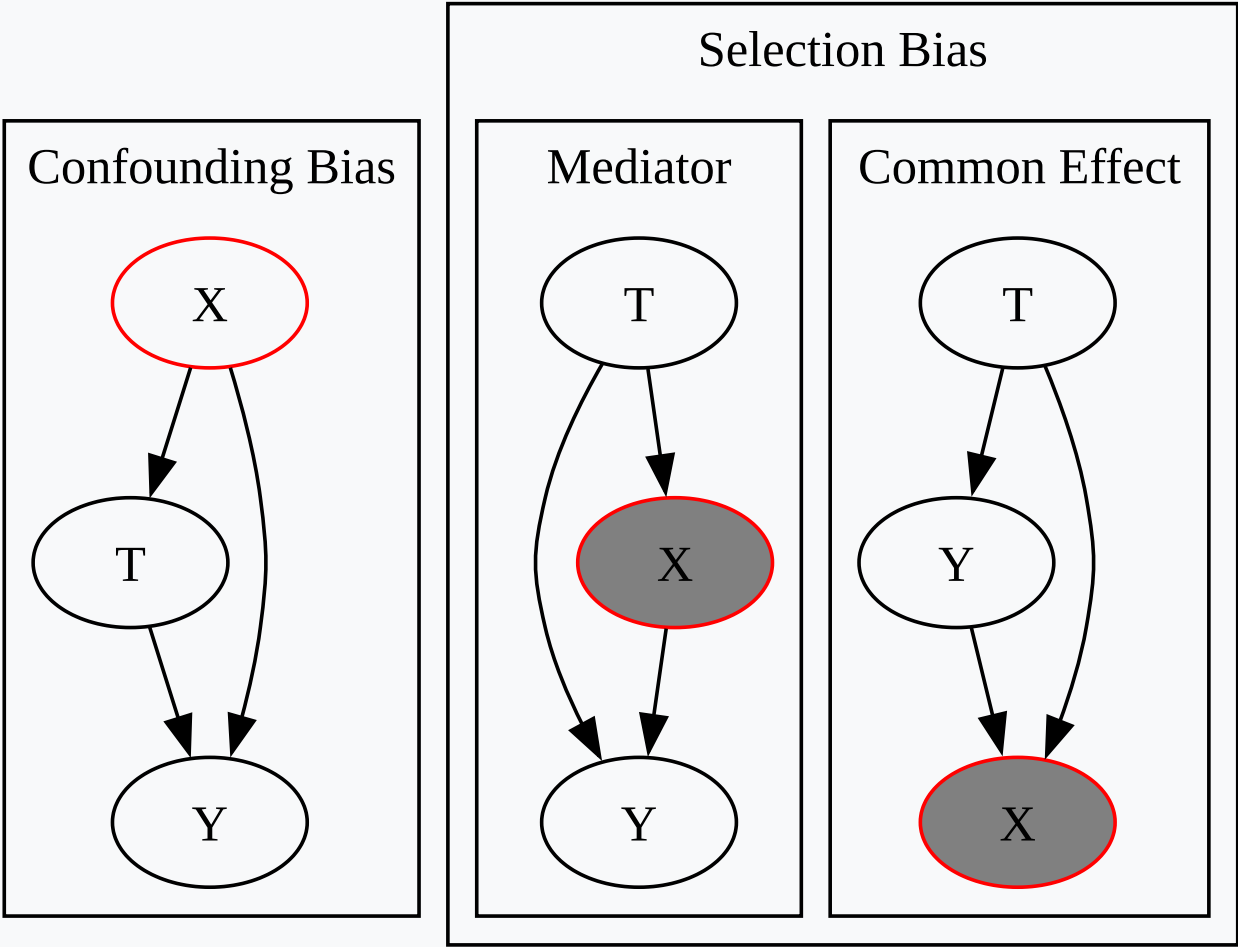
`study hours` is a **mediator**: it lies on the path from the treatment to the outcome. Conditioning on it **closes that path**, and the measured effect shrinks.

Interested students study more and perform better — but among people who all study a lot, even the uninterested ones do well, so `interest` and `performance` look barely related.



Main Sources of Bias

Main Sources of Bias



Confounders

We **should** control on the common cause

Selection bias — common effect

We should **not** control on the common effect

Selection bias — mediator

We should **not** control on the mediator



Linear Regression for Causal Analysis

"Controlling for X " so far meant splitting the data into groups and measuring the association inside each. **Linear regression automates it:** a variable we include among the predictors is a variable we have controlled for.

Ignoring the confounder

$$performance = \beta_0 + \beta_1 tutoring$$

	coef	std err	t	P> t
Intercept	72.9291	1.199	60.832	0.000
tutoring	6.8269	1.671	4.087	0.000

Students who enroll increase their average performance by **6.83 points** — on a 0–100 scale, and significant.

The intercept of the first model is the average performance of the untreated; the coefficient of **tutoring** is the average difference between the groups. Regression also gives us confidence intervals and tests for free.

Controlling for `interest`

$$performance = \beta_0 + \beta_1 tutoring + \beta_2 interest$$

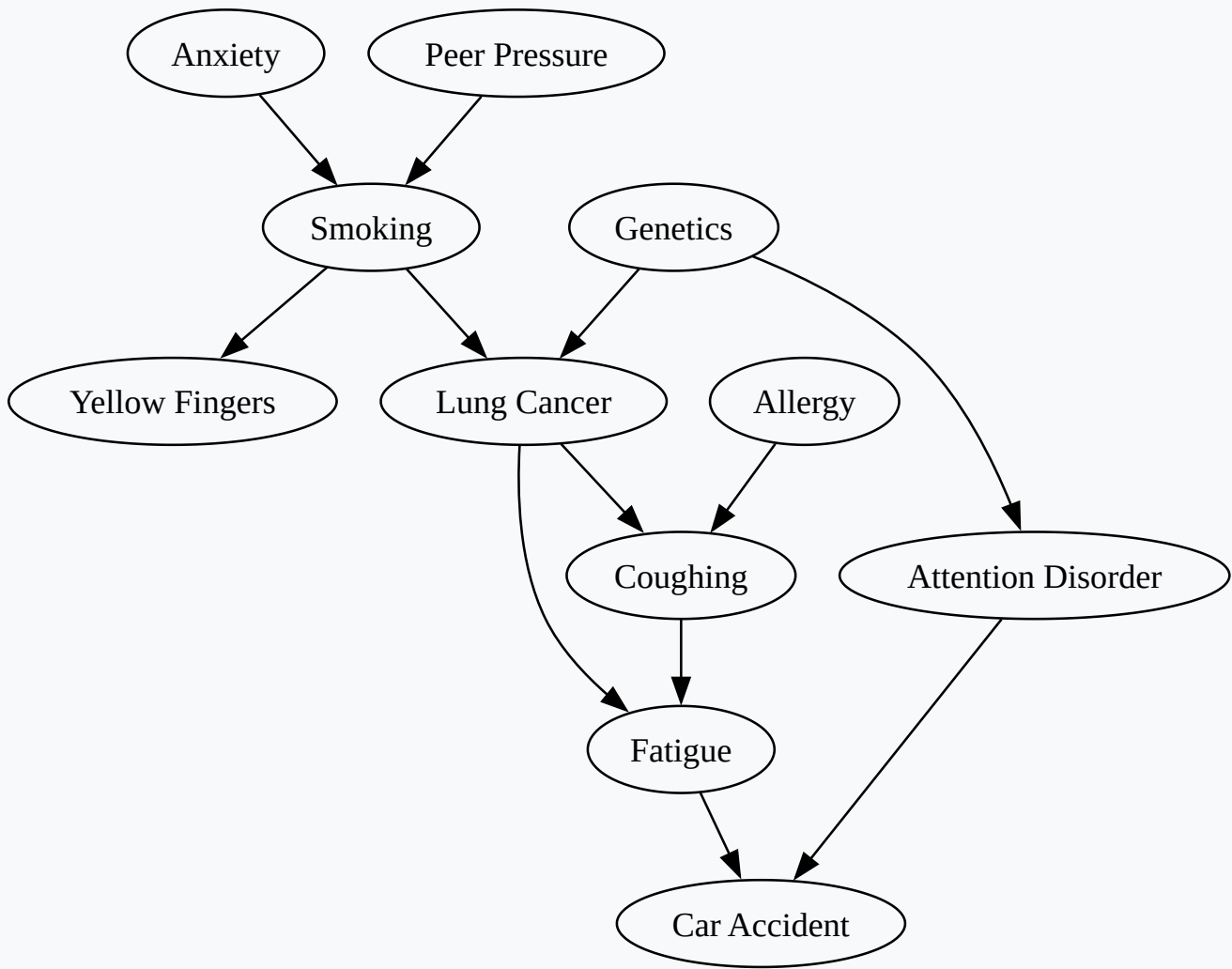
	coef	std err	t	P> t
Intercept	54.1697	1.296	41.804	0.000
tutoring	-0.9516	1.129	-0.843	0.400
interest	7.8097	0.441	17.711	0.000

With **interest** held constant the effect of tutoring is **not distinguishable from zero**.



Example on the LUCAS Dataset

LUCAS is an **artificial dataset**, generated from a known causal graph — which means we can check our reasoning against the truth. Does **Smoking** cause **Car Accident** ?



Smoking alone

	coef	P> z
Intercept	0.7668	0.000
Smoking	0.2596	0.022

$e^{0.2596} \approx 1.3$: smoking is associated with a **+30%** increase in the odds of a car accident. Is that a causal effect?

Three paths connect the two nodes. One is **blocked by a collider** (**Lung cancer** , not conditioned on). The other two are open: smoking can reach **Car Accident** directly through **Fatigue** , or through **Coughing** . Conditioning on **Coughing** closes the second:

	coef	P> z
Intercept	0.3411	0.002
Smoking	-0.0130	0.914
Coughing	0.9529	0.000

Once the influence of **Coughing** is removed, the coefficient of **Smoking** is **not significantly different from zero**: there is no significant causal effect between the two variables.



Conclusions and Next Steps



We Have Explored:

Potential outcomes and the treatment effect

Where the bias in a correlation comes from

Randomised controlled trials

Graphical causal models

Chains, forks and colliders

Confounding and selection bias

Regression as an adjustment device

- This closes the first block. In the next lectures we change goal: from explaining a relationship to predicting an outcome.

References

- Facure Alves, Matheus. *Causal Inference for the Brave and True*. matheusfacure.github.io/python-causality-handbook
- Pearl, Judea, and Dana Mackenzie. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.
- The LUCAS dataset: causality.inf.ethz.ch/data/LUCAS.html
- Lecture notes: [Chapter 11 — Causal analysis](#)

